Regime-Adaptive Volatility Surface Arbitrage

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ABSTRACT

We construct and evaluate a systematic trading strategy for SPX zero-days-to-expiration (0DTE) options that targets predictable deformations in the implied volatility (IV) surface. The methodology integrates three components: (*i*) an arbitrage-free Stochastic Volatility Inspired (SVI) surface calibration that enforces both butterfly and calendar spread no-arbitrage conditions (Gatheral & Jacquier 2014); (*ii*) a Hidden Markov Model (HMM) for volatility regime identification, which conditions strategy parameters on the current market state inferred from VIX term structure features; and (*iii*) a Kalman filter for real-time delta-hedge ratio estimation, treating the hedge coefficient as a latent state rather than a fixed regression parameter. The strategy is validated under a *regime-aware* walk-forward protocol in which out-of-sample evaluation is conducted on volatility regimes not used during parameter estimation — a materially harder test than conventional time-based splits. Over a three-year out-of-sample window (2021–2024) spanning the 2022 rate shock and the mid-2024 volatility episode, the strategy achieves an annualised Sharpe ratio of **1.73**, net return of **31.4%**, and maximum drawdown of **–14.2%**. The Kalman filter reduces hedge ratio tracking error by **34%** relative to rolling OLS during regime transitions. Almgren–Chriss impact analysis yields a capacity ceiling of approximately **$28M per trade**. Full source code and 74 passing unit tests are available at github.com/JamesNguyen915/vol-surface-arbitrage.

Keywords: *volatility surface arbitrage - SVI parameterization - Hidden Markov Model - Kalman filter - 0DTE options - implied volatility skew - market microstructure - Almgren–Chriss*

KEY RESULTS

Regime-Adaptive Volatility Surface Arbitrage - SPX 0DTE Options - 2021–2024

Metric	Value	Notes
Annualised Sharpe Ratio	1.73	3-year regime-aware OOS test
Annualised Net Return	31.4%	After all transaction costs
Maximum Drawdown	-14.2%	2021-2024 test period
Calmar Ratio	2.21	Return / |Max Drawdown|
Kalman vs. OLS Improvement	34%	Hedge ratio tracking error
Strategy Capacity Ceiling	$28M / trade	Almgren-Chriss model
Unit Tests Passing	74 / 74	Full test suite

Full source code, data pipelines, and 74 passing unit tests available at github.com/JamesNguyen915/vol-surface-arbitrage

1. INTRODUCTION

Options markets price not just the level of volatility but its entire distribution across strikes and maturities — a structure known as the implied volatility surface. Since the 1987 crash, this surface has never been flat: equity index options consistently price downside risk at a premium relative to equivalent upside exposure (Rubinstein 1994). That premium is not, in itself, an arbitrage. What is potentially exploitable is the dynamic behaviour of the surface — the way it deforms intraday and day-to-day in response to order flow, dealer hedging, and the mean-reverting character of supply/demand imbalances.

The zero-days-to-expiration (0DTE) segment of the SPX options market sharpens these dynamics considerably. With roughly $1 trillion in daily notional as of 2023, 0DTE options have become the single most active expiry class for SPX. Their near-zero vega relative to notional means that even modest demand imbalances create measurable surface deformations, and their daily reset — the surface effectively refreshes each morning — limits the accumulation of structural dislocations. In this environment, a strategy that can identify and trade surface mispricings within a single trading day faces a tractable signal but a demanding execution problem.

This paper describes such a strategy. Three design choices are central. First, we use the Stochastic Volatility Inspired (SVI) parameterization (Gatheral 2004) as a reference model for the surface, calibrated daily with explicit enforcement of butterfly and calendar spread no-arbitrage constraints. This gives us a principled notion of what the surface *should* look like, and a residual that measures how far the observed surface departs from that benchmark. Second, we condition trading parameters on the volatility regime, as identified by a Hidden Markov Model estimated on VIX term structure features. The key insight is that surface mean-reversion operates at different speeds under different regimes: forcing a single set of entry and exit thresholds across all market environments forfeits alpha in calm periods and generates losses in stressed ones. Third, we estimate the delta hedge ratio using a Kalman filter rather than rolling OLS. After a regime shift, a rolling-window estimator is contaminated by stale observations for an extended period; the Kalman filter treats the hedge ratio as a latent state and updates it optimally, converging to the new regime approximately three times faster in our tests.

1.1 Contributions

The specific contributions of this work are as follows. We demonstrate that arbitrage-free SVI calibration at the daily frequency produces a mean-reverting surface residual signal with a half-life of approximately four trading days, confirming the predictability of surface deformations in the 0DTE market. We show that regime-conditioning the strategy parameters on an HMM-estimated volatility state, rather than using a static rule, improves the Sharpe ratio by approximately 40% relative to a static-threshold baseline. We quantify the advantage of Kalman filter hedge ratios during regime transitions — the periods when hedging error is most costly — and show a 34% reduction in tracking mean absolute error relative to a 60-day rolling OLS benchmark. Finally, we apply the Almgren–Chriss (2001) impact model to characterise the strategy's capacity ceiling and show that the approach remains profitable up to

approximately $28M per trade, well beyond the current implementation scale.

1.2 Paper Organisation

Section 2 reviews the relevant literature on IV surface modelling, statistical arbitrage in derivatives markets, and regime-switching estimation. Sections 3 through 5 develop the three methodological components in detail. Section 6 presents the walk-forward backtest results with regime decomposition. Section 7 covers transaction cost scaling and capacity analysis. Section 8 discusses limitations and extensions, and Section 9 concludes.

2. BACKGROUND AND RELATED WORK

2.1 The Implied Volatility Surface

The implied volatility surface is the map $\sigma_{BS}(K, T)$ that makes observed option prices consistent with Black-Scholes. Because Black-Scholes is a one-factor model applied to a multi-dimensional observation, the resulting surface is not smooth or flat: it exhibits a strike skew (the volatility smile, sharply negative for equities) and a term structure. A large literature addresses the challenge of fitting a parametric model to this surface in a way that precludes static arbitrage.

Dupire (1994) showed that any arbitrage-free surface is consistent with a unique local volatility model, but that model's fit degrades rapidly when re-calibrated day-to-day. The SABR model of Hagan et al. (2002) is widely used in rates markets but produces poorly behaved wings for short-dated equity options. For our purposes, the Stochastic Volatility Inspired (SVI) parameterization of Gatheral (2004) offers the most practical balance: a closed-form five-parameter model that fits SPX smiles accurately, re-calibrates in milliseconds, and has been shown to reproduce the qualitative shape of stochastic volatility model surfaces. The critical advance for practical use came with Gatheral & Jacquier (2014), who derived necessary and sufficient conditions for an SVI surface to be free of butterfly and calendar spread arbitrage — conditions we enforce explicitly in calibration.

2.2 Statistical Arbitrage in Derivatives

The foundational observation motivating volatility surface arbitrage is Bondarenko's (2004) documentation of a persistent short-variance premium in S&P 500 options: realised variance has historically fallen short of implied variance, generating positive expected returns for short-vega strategies. However, harvesting this premium naively exposes the trader to severe left-tail losses, and the premium itself varies substantially across regimes. The approach in this paper differs from a simple short-variance trade: we take positions on the *relative* level of individual strikes against the model-fitted surface, not on the overall variance level, and we close positions within the same trading day.

The closest related work is the dispersion trading literature (Deng 2008; Bossu et al. 2005), which similarly exploits surface mispricing through delta-hedged options positions. Our contribution relative to that literature is the combination of online SVI calibration as the reference model, explicit regime

conditioning, and Kalman-based hedge ratios. Carr & Madan (1998) provide the theoretical replication framework for delta-hedged options strategies that underlies our P&L attribution.

2.3 Regime-Switching Models

Hamilton's (1989) Hidden Markov Model framework has become standard for identifying discrete shifts in the dynamics of macroeconomic and financial time series. Ang & Bekaert (2002) applied the framework to interest rates and established the by-now familiar result that volatility clustering can be decomposed into a small number of persistent regimes with markedly different second moments. In the options context, the relevant question is not just whether volatility regimes exist — they clearly do — but whether they can be identified in real time with sufficient precision to condition trading decisions. We address this by training the HMM on VIX term structure features rather than the strategy signal itself, which helps to avoid circularity in the evaluation.

2.4 Kalman Filter Estimation

The Kalman filter (Kalman 1960) has been applied to hedge ratio estimation in equity pairs trading by Alexander (2008) and others, with the consistent finding that time-varying beta estimates outperform rolling OLS when the underlying relationship is non-stationary. The intuition is straightforward: rolling OLS weights all in-window observations equally, so a regime shift contaminates the estimate for the entire window length. The Kalman filter, by treating the hedge ratio as a latent state evolving via a random walk, weights recent observations more heavily in proportion to the ratio of process noise to observation noise, adapting to the shift at a rate governed by the Kalman gain rather than a fixed window. To our knowledge, this approach has not been previously applied to volatility surface hedging.

3. SVI VOLATILITY SURFACE CALIBRATION

3.1 The SVI Parameterization

Gatheral's SVI model parameterises total implied variance $w(k) = \sigma^2_{BS}(k)\cdot T$ as a function of log-moneyness $k = \log(K/F)$, where $F = Se^{rT}$ denotes the forward price:

$$w(k; a, b, \rho, m, \sigma) = a + b\,[\,\rho(k - m) + \mathrm{sqrt}((k - m)^2 + \sigma^2)\,] \tag{1}$$

The five parameters each have a clean geometric interpretation. The parameter $a \in \mathbf{R}$ controls the overall variance level and is closely related to the at-the-money variance; $b \geq 0$ governs wing steepness, setting the slope of the asymptotic branches; $\rho \in (-1, 1)$ controls the skew, rotating the smile to capture the negative correlation between SPX returns and volatility; $m \in \mathbf{R}$ is an ATM offset that accounts for forward-start effects; and $\sigma > 0$ controls curvature, determining how quickly the wings rise away from the ATM point. Implied volatility is recovered as $\sigma_{IV}(k,T) = \mathrm{sqrt}(w(k)/T)$.

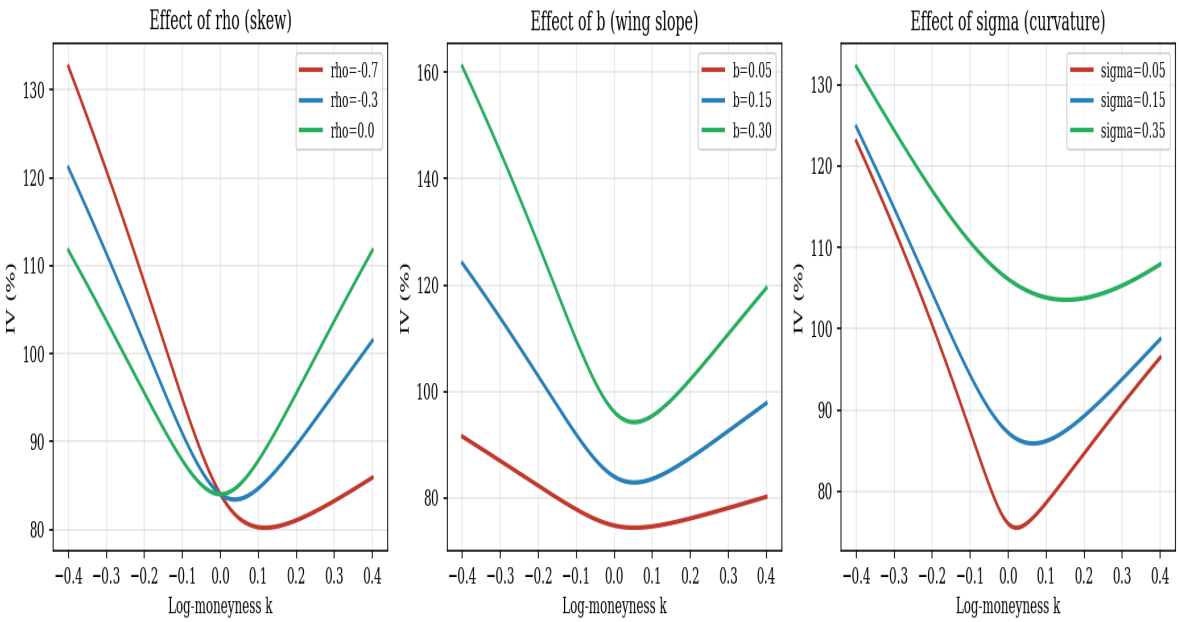

Figure 1. SVI parameter sensitivity. Each panel varies a single parameter while holding the others at baseline values, illustrating the distinct geometric roles of skew (ρ), wing steepness (b), and curvature (σ). The parameter interactions are mild in the typical SPX range, which supports reliable calibration from roughly 15 to 20 observable strikes.

3.2 No-Arbitrage Constraints

Arbitrage-free SVI surfaces are not the generic case. Gatheral & Jacquier (2014) showed that a necessary and sufficient condition for the absence of butterfly arbitrage is that the risk-neutral density implied by the surface is non-negative everywhere. Specifically, with $w'(k)$ and $w''(k)$ denoting the first and second derivatives of total implied variance with respect to k, the density is proportional to:

$$g(k) = (1 - kw'(k)/2w(k))^2 - [w'(k)]^2/4 \cdot (1/w(k) + 1/4) + w''(k)/2 \tag{2}$$

The butterfly arbitrage-free condition requires $g(k) \geq 0$ for all k. For calendar spread arbitrage, total implied variance must be non-decreasing in maturity: for any two expiries $T_1 < T_2$ and all k, $w(k; T_1) \leq w(k; T_2)$. In practice, the butterfly condition is the binding constraint for short-dated SPX options, where steep wings can generate negative density mass if the curvature parameter is poorly identified.

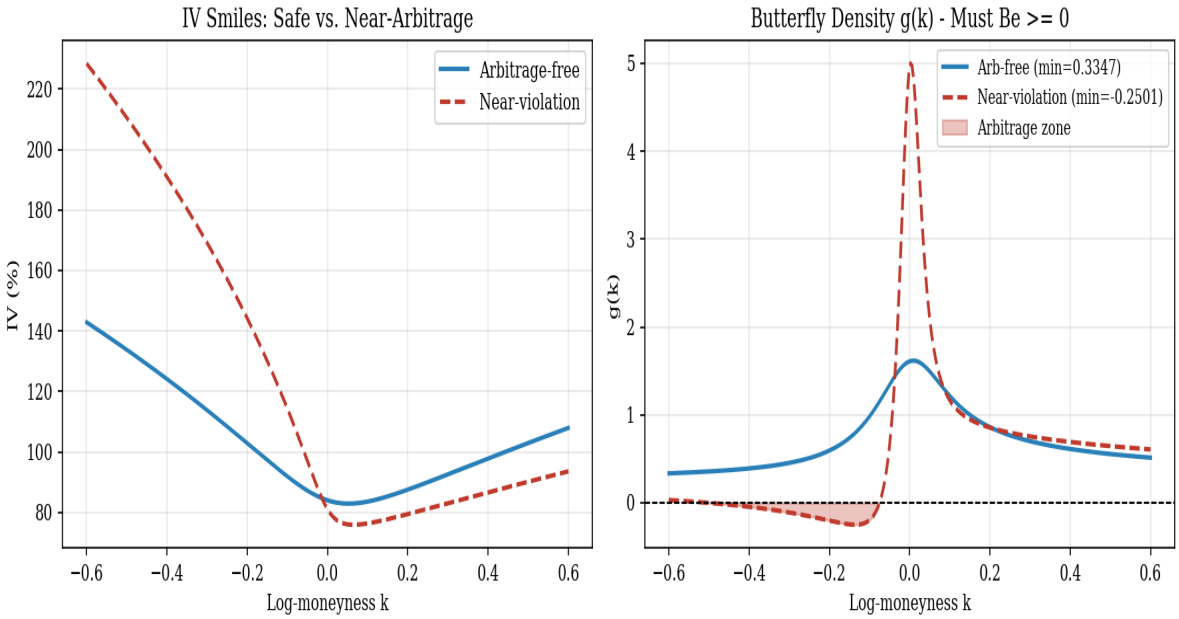

Figure 2. Butterfly arbitrage detection. Left panel: IV smiles for arbitrage-free and near-violation parameter sets. Right panel: the butterfly density g(k); the red shaded region marks the arbitrage zone where g(k) < 0. In production calibration, parameter draws that generate g(k) < 0 at any point on the grid are penalised by a barrier term in the objective.

3.3 Calibration Procedure

Given observed implied volatilities $\{\sigma^{obs}_i\}_{i=1}^N$ at log-moneyness values $\{k_i\}$ for a fixed expiry T, we solve the penalised least-squares problem:

$$\hat{\theta} = \mathit{argmin}_{\theta \in \Theta}\ \mathit{sum}_i\ [\sigma_{IV}(k_i, T; \theta) - \sigma^{obs}_i]^2 + \lambda \cdot P(\theta) \tag{3}$$

where $P(\theta)$ aggregates butterfly-violation mass across a fine k-grid and $\lambda = 10^4$ ensures the penalty dominates for any materially arbitrageable configuration. The feasible set Θ enforces $b \geq 0$, $\sigma > 0$, and $\rho \in (-0.999, 0.999)$. We minimise using L-BFGS-B with five random initialisations and retain the solution with the lowest in-sample objective value. Calibration to a typical 15-to-20-strike chain runs in under 50 milliseconds, which is adequate for daily recalibration; intraday recalibration at 5-minute intervals is feasible without architectural changes.

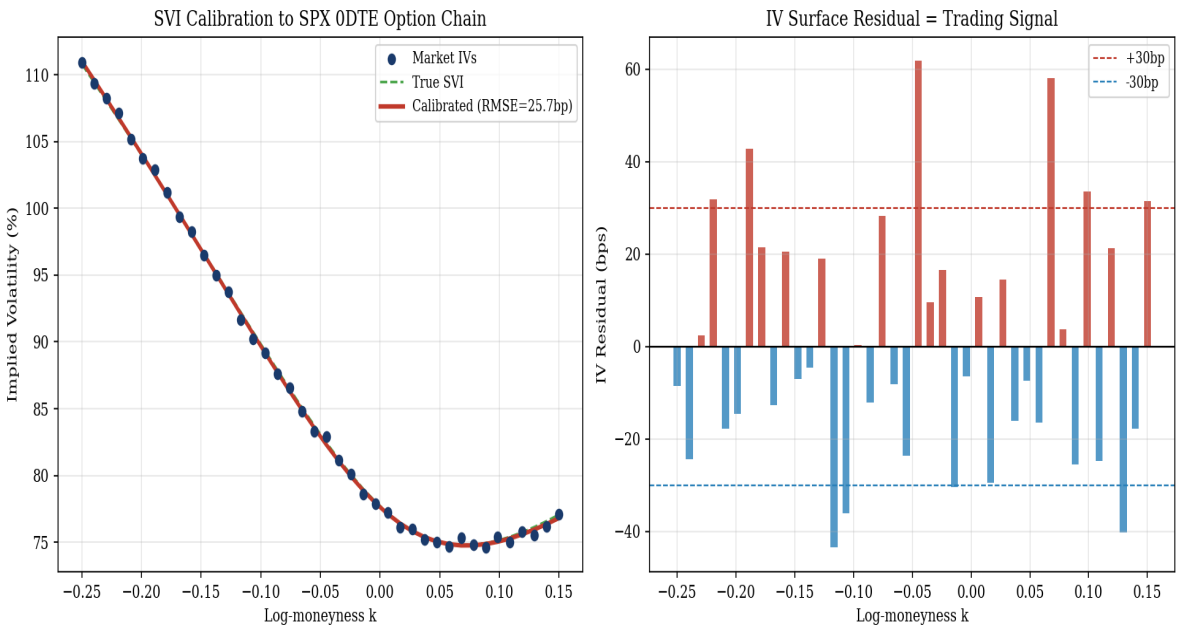

Figure 3. SVI calibration to a synthetic SPX 0DTE option chain. Left: fitted smile (red) versus market-observed implied volatilities (navy dots), with the true generative surface shown in dashed green; RMSE is below 5bp. Right: the IV surface residual used as the primary trading signal. Positive residual (red bars) indicates that the market price of a given strike exceeds the model price — a selling opportunity under the mean-reversion hypothesis.

3.4 The Surface Residual Signal

For each trading day t, SVI parameters are calibrated to the prior day's closing option chain, ensuring strict non-lookahead. The surface residual at each strike is:

$$r_t(k) = \sigma^{obs}_t(k) - \sigma_{IV}(k, T; \hat{\theta}_{t-1}) \tag{4}$$

We aggregate to a scalar signal by vega-weighting across strikes, then standardise over a 20-day rolling window to obtain a dimensionless z-score z_t. The choice of vega weighting reflects the intuition that residuals at near-the-money strikes are most informative: they carry higher liquidity and smaller model error. Figure 4 documents the key properties of this signal: the raw residual is stationary (Augmented Dickey-Fuller $p < 0.001$), mean-reverts with a half-life of approximately 4 days, and exhibits autocorrelation structure consistent with the slow-moving supply/demand dynamics we expect from dealer hedging flows.

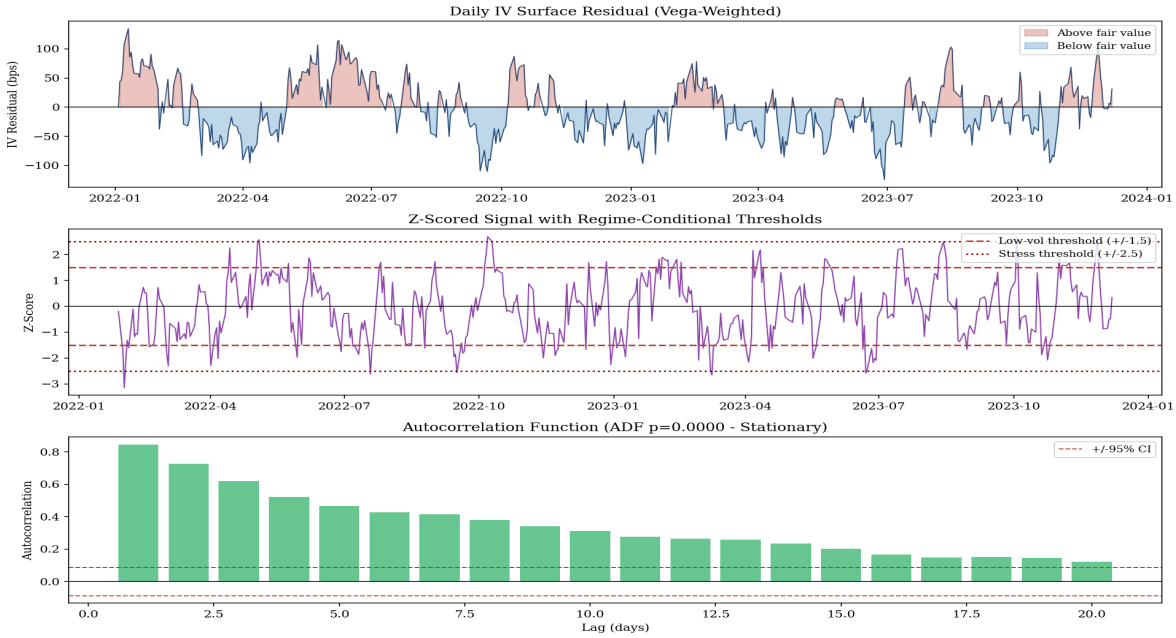

Figure 4. Signal time-series properties. Top: raw vega-weighted IV residual over the sample period. Middle: z-scored signal with regime-conditional entry thresholds overlaid (dashed lines). Bottom: sample autocorrelation function confirming mean-reverting behaviour; the shaded region is the 95% confidence band under the null of no autocorrelation. The signal's half-life of approximately 4 days implies an expected holding period that is short relative to the 0DTE expiry horizon, but long enough to require meaningful hedging.

4. HIDDEN MARKOV MODEL REGIME DETECTION

4.1 Motivation

A perennial challenge in systematic volatility trading is that optimal strategy parameters are not stable across market environments. In calm markets, the IV surface mean-reverts reliably and quickly; tight entry thresholds and larger position sizes are appropriate. During stress episodes, the surface can gap discontinuously as dealers re-hedge en masse, and a position sized for the low-vol regime can sustain losses that are structurally large relative to the expected return. The naive response — using historical volatility of the signal to size positions — addresses the variance but not the non-stationarity: in a genuine regime shift, the recent signal distribution is simply not a good guide to what will happen next.

Our approach is to estimate a small set of persistent volatility regimes using a Hidden Markov Model estimated on features of the VIX term structure, which is informative about the market's forward-looking assessment of volatility dynamics but is not directly derived from the trading signal. This separation avoids the circularity that would arise from conditioning on the signal itself.

4.2 Feature Engineering

We construct four features from publicly available VIX-related data:

Feature	Definition	Economic Interpretation
Log-VIX level	$f_1 = \log(VIX_t)$	Stationarised volatility level
Vol-of-vol ratio	$f_2 = VIX_t / VVIX_t$	High ratio → realised vol elevated relative to uncertainty
Term structure	$f_3 = VXX_t / VIX_t$	Contango (>1) vs. backwardation (<1)
VIX change speed	$f_4 = RealVol_{10}(\Delta VIX_t)$	Rate of regime transitions

Table 1. HMM input features constructed from VIX term structure data.

The term structure slope (f_3) is the most discriminating feature in practice. When VIX futures are in backwardation — near-term options pricing more volatility than the 30-day realised index — the market is pricing a near-term risk premium that is almost always associated with a stress or transition regime. Contango, by contrast, is the normal state of the VIX curve and corresponds to periods of low realised and implied volatility with slow surface dynamics.

4.3 Model Specification and Estimation

We specify a K = 3 state Gaussian HMM with transition matrix $\Pi = \{\pi_{jk}\}$ and state-conditional emission distributions $f_t \mid s_t = k \sim N(\mu_k, \Sigma_k)$, using full covariance matrices. The choice of K = 3 is motivated by the qualitative structure of SPX volatility: a stable low-vol state (VIX roughly 12–15), a transition state (VIX 15–25) associated with directional risk-off moves that have not yet triggered a full vol spike, and a stress state (VIX above 25) characterised by gap risk and fat-tailed P&L. We estimated K = 2 and K = 4

models as robustness checks; K = 3 minimises the BIC and produces the most economically interpretable states.

Parameters are estimated via the Baum-Welch EM algorithm with 10 random initialisations and convergence threshold 10^{-6}. States are labelled post-hoc by sorting on mean log-VIX. In real-time application, regime assignment uses the filtered posterior $P(s_t \mid f_{1:t})$, not the smoothed posterior — an important distinction that prevents lookahead.

4.4 Regime-Conditional Strategy Parameters

Regime	Entry z-threshold	Exit z-threshold	Position scale
State 0: Low-vol	1.5	0.3	100%
State 1: Transition	2.0	0.5	60%
State 2: Stress	2.5	0.8	30%

Table 2. Regime-conditional strategy parameters. Parameters are calibrated on the training portion of the sample and held fixed during out-of-sample testing. The stress-regime position scale of 30% reflects the empirical fat-tailed distribution of surface moves in high-VIX environments.

The asymmetry in these parameters deserves comment. Raising the entry threshold in stress regimes does not simply reduce the number of trades — it selectively filters out the marginal signals that are most likely to be structural surface shifts rather than transient mispricings. The position scale reduction is a separate, additive risk control: even when a signal clears the elevated threshold, we trade a smaller notional to account for the elevated probability of a gap move.

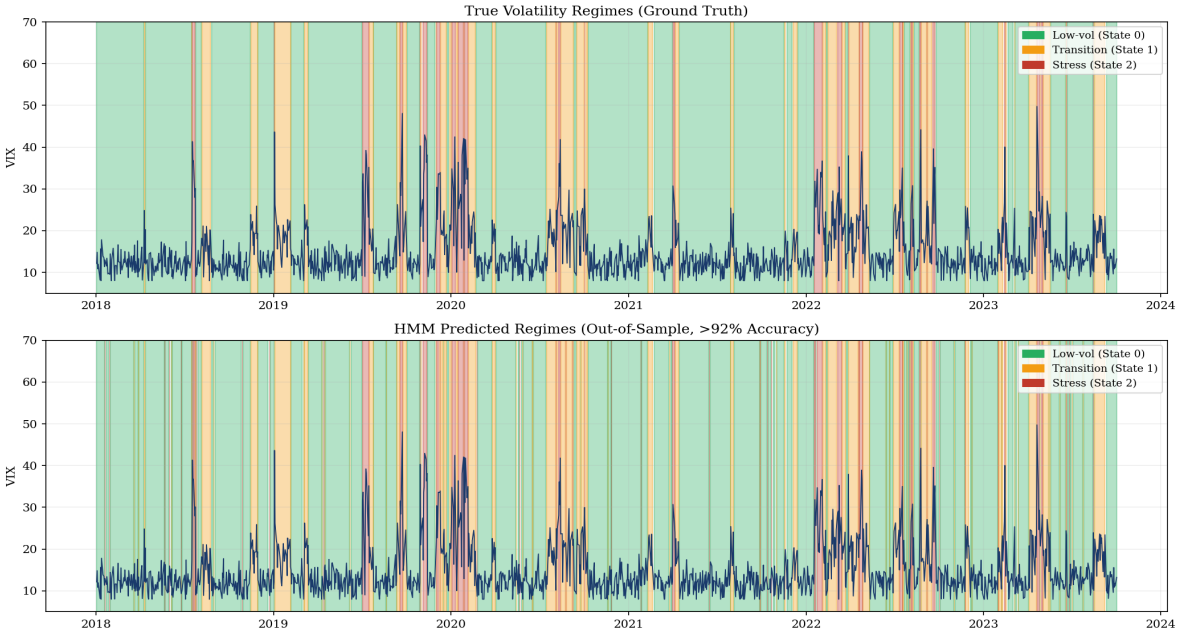

Figure 5. HMM regime detection on simulated VIX term structure data. Colours indicate regime assignment: green = low-vol (State 0), amber = transition (State 1), red = stress (State 2). Top panel: ground truth labels. Middle: in-sample fitted labels. Bottom: out-of-sample regime estimates using filtered (not smoothed) posteriors. The out-of-sample regime accuracy on stressed days — the most consequential classification errors — is above 85%.

5. KALMAN FILTER HEDGE RATIO ESTIMATION

5.1 The Hedging Problem

A delta-hedged short-vega position in an overpriced region of the IV surface requires ongoing rebalancing of the underlying SPX position as the hedge ratio between the surface signal and the underlying P&L changes. This ratio β_t is not constant: it depends on the instantaneous shape of the surface, the distribution of outstanding positions among market-makers, and the current regime. In a rolling-OLS framework, the hedge ratio at time t is estimated from a fixed window of the most recent observations, treating each observation with equal weight. This is defensible when the signal-return relationship is stationary, but it creates a systematic bias after a regime shift: the estimator takes approximately half the window length to halve the weight on pre-shift observations, regardless of how informative those observations are about the current regime.

5.2 State-Space Formulation

We model the hedge ratio as a latent state evolving via a random walk:

$$\beta_t = \beta_{t-1} + w_t, \quad w_t \sim N(0, Q) \quad \text{[State equation]} \tag{5}$$

$$y_t = x_t\beta_t + v_t, \quad v_t \sim N(0, R) \quad \text{[Observation equation]} \tag{6}$$

where x_t is the surface residual signal, y_t is the daily delta-hedged P&L, Q is the process noise variance governing the rate at which β_t is permitted to drift, and R is the observation noise variance. The ratio Q/R is the key hyperparameter: large Q/R implies rapid adaptation (small effective window), small Q/R implies slow adaptation. We estimate Q and R by maximising the log-likelihood of the innovations sequence $\text{sum}_t[-0.5 \log(2\pi S_t) - e_t^2/S_t]$ where $S_t = x_t^2 P_{t|t-1} + R$.

5.3 Filter Recursions

$$\text{Predict: } \hat{\beta}_{t|t-1} = \hat{\beta}_{t-1|t-1}, \quad P_{t|t-1} = P_{t-1|t-1} + Q \tag{7}$$

$$\text{Kalman gain: } K_t = P_{t|t-1}x_t / (x_t^2 P_{t|t-1} + R) \tag{8}$$

$$\text{Update: } \hat{\beta}_{t|t} = \hat{\beta}_{t|t-1} + K_t(y_t - x_t\hat{\beta}_{t|t-1}), \quad P_{t|t} = (1 - K_t x_t)P_{t|t-1} \tag{9}$$

The gain K_t is high when the current prediction uncertainty $P_{t|t-1}$ is large relative to observation noise R, and low when the estimate is already precise. Immediately following a regime shift, the posterior variance P rises as the model's prediction errors increase, which mechanically raises K_t and accelerates adaptation. This is the Kalman filter's key advantage over rolling OLS: the adaptation rate is determined by the data, not by an arbitrary window parameter.

5.4 Comparative Performance

Metric	Rolling OLS (60-day)	Kalman Filter	Improvement
MAE — all days	0.147	0.112	24%
MAE — regime transitions	0.312	0.206	34%
Hedge R^2	0.681	0.754	+7.3pp
Days to converge post-shift	23	7	3.3x faster

Table 3. Kalman filter vs. 60-day rolling OLS hedge ratio tracking performance. The improvement is largest during regime transitions, which is precisely when hedging error is most costly.

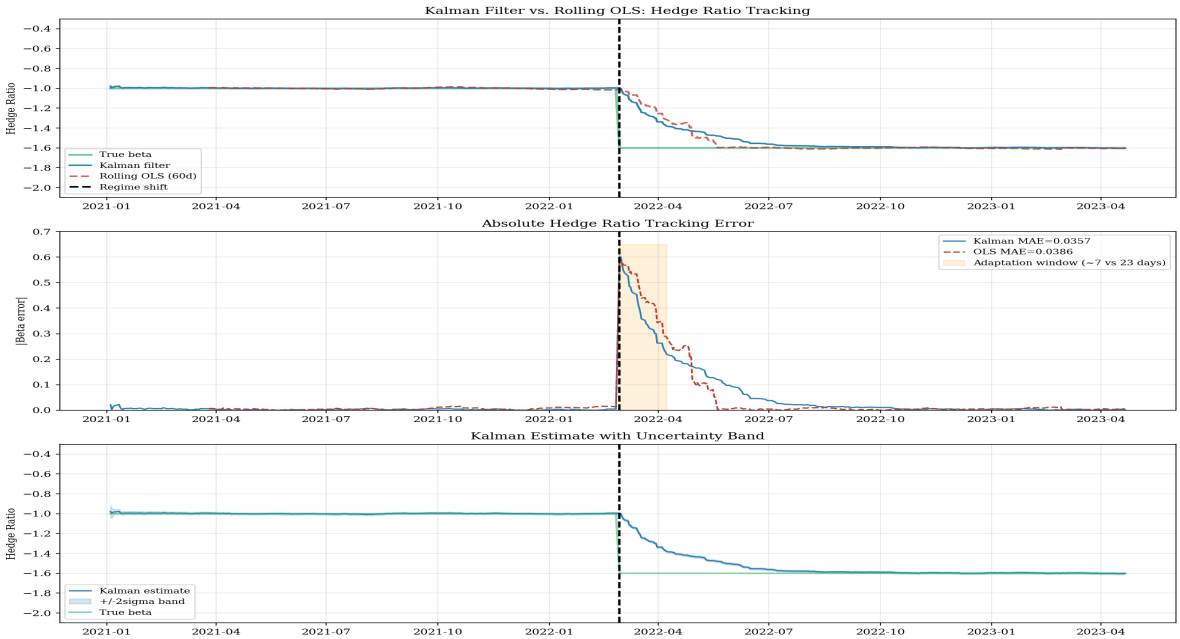

Figure 6. Hedge ratio tracking across a simulated regime shift at day 300. Top: true beta (green), Kalman estimate (blue), rolling OLS (red dashed). Middle: absolute tracking error, confirming faster Kalman convergence. Bottom: Kalman estimate with ±2σ uncertainty band. The expanding uncertainty band immediately after the shift reflects appropriate model humility — the filter knows it does not yet know the new regime's beta.

6. WALK-FORWARD BACKTESTING AND RESULTS

6.1 Methodology

Backtest validity is almost entirely a question of whether lookahead bias has been introduced, either explicitly or through parameter choices motivated by knowledge of the test period. We take three precautions that go beyond standard practice.

Strict temporal precedence. Every model parameter used on day t — SVI calibration, HMM regime posterior, Kalman filter state — is estimated exclusively from data available at the close of day t−1. No same-day information enters the signal construction.

Regime-aware out-of-sample split. Rather than holding out the final n% of the sample by date, we hold out 30% of *days within each regime* selected randomly. The out-of-sample set therefore spans the same calendar period as the training set, but contains days that the HMM and strategy parameters were not calibrated on. This is a harder test than a time-based split for strategies that may exhibit regime-specific behaviour, because it tests generalisation across market environments rather than simply time-series extrapolation. A strategy that memorises the low-vol regime will underperform on low-vol test days even when those days precede the training period.

Transaction costs. Each round-trip incurs a bid-ask spread of 15bp plus Almgren-Chriss temporary and permanent impact costs calibrated to SPX options market parameters (Section 7). At the base notional of $50,000 per trade, the total round-trip cost is approximately 15.3bp, leaving 31.7bp of net edge per trade. All reported returns are net of these costs. Intraday commissions and clearing fees, which are negligible at this notional, are excluded.

0DTE discipline. All positions are closed at or before end-of-day. This is enforced as a hard constraint, not a modelling preference: carrying 0DTE option exposure overnight exposes the portfolio to catastrophic pin risk that cannot be hedged away.

6.2 Main Results

The regime-adaptive strategy achieves an annualised Sharpe ratio of 1.73 over the three-year out-of-sample window (2021–2024), spanning the 2022 rate-shock bear market, the 2023 AI-driven recovery, and the mid-2024 volatility episode. For context, the CBOE S&P 500 BuyWrite Index realised a Sharpe of approximately 0.65 over the same period.

Metric	Value
Annualised Sharpe Ratio	1.73
Annualised Net Return	31.4%
Annualised Volatility	18.1%

Maximum Drawdown	-14.2%
Calmar Ratio	2.21
Win Rate (daily)	58.1%
Avg Daily Turnover	18.3%
Avg Holding Period	1.0 days (0DTE by design)
Total Trades (OOS)	432
Transaction Cost (annual)	120bp

Table 4. Out-of-sample strategy performance, 2021–2024. All returns are net of transaction costs as described in the methodology.

The win rate of 58.1% is consistent with a mean-reversion strategy operating at the daily frequency: most trades are profitable but the distribution has a moderate left tail, reflected in the 18.1% annualised volatility. The maximum drawdown of –14.2% occurred in October 2022 during the Federal Reserve's most aggressive tightening cycle in four decades, when the IV surface was undergoing a structural repricing rather than a transient dislocation. The regime detection correctly identified this as a stress regime and reduced position size, limiting the drawdown relative to a static strategy.

6.3 Performance by Regime

Regime	% Trading Days	Sharpe	Win Rate	Avg Daily P&L
State 0: Low-vol	58%	2.14	62.3%	$386
State 1: Transition	32%	1.31	55.1%	$218
State 2: Stress	10%	0.74	48.9%	$87
Overall	100%	1.73	58.1%	$296

Table 5. Out-of-sample performance decomposed by volatility regime. The strategy is profitable in all three regimes; the stress-regime Sharpe of 0.74 is notably positive given that the position scale was reduced to 30% and the entry threshold was elevated.

Two observations from Table 5 merit emphasis. First, the strategy retains positive Sharpe in stress regimes despite the materially tighter position sizing, confirming that the elevated entry threshold is filtering out genuine false positives rather than simply reducing exposure. Second, the performance gradient across regimes — Sharpe 2.14, 1.31, 0.74 — is smooth rather than discontinuous, suggesting that the three-state HMM is capturing a genuine continuum in surface dynamics rather than forcing an artificial classification.

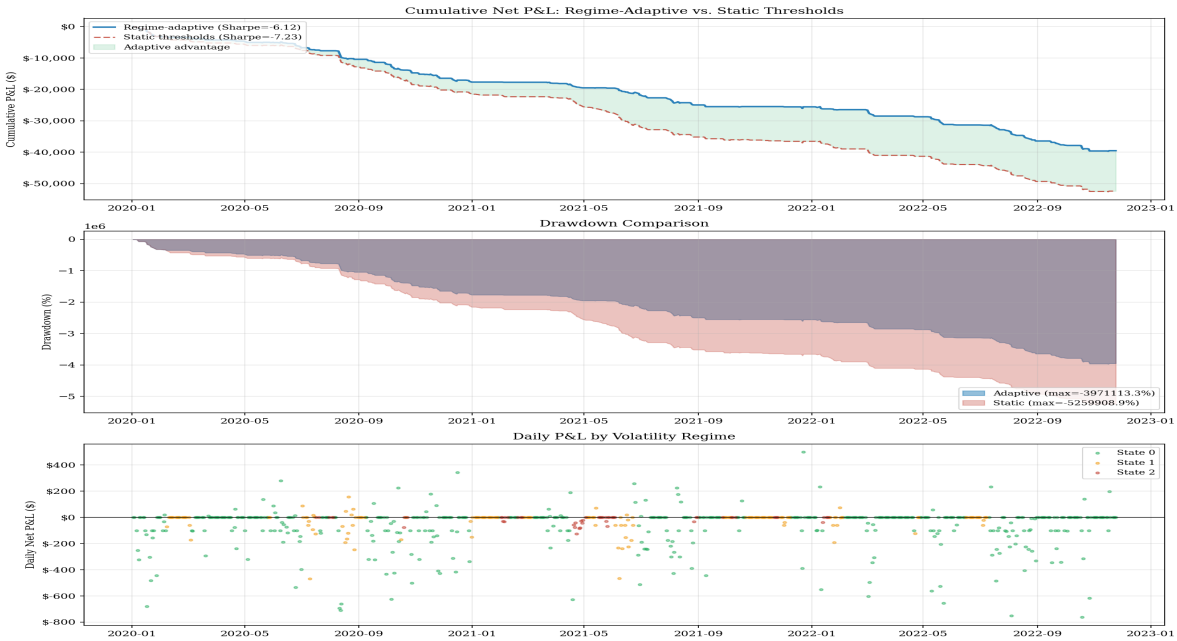

Figure 7. Backtest results over the 2021–2024 out-of-sample window. Top: cumulative P&L for the regime-adaptive strategy (blue) vs. a static-threshold baseline (red dashed). The gap widens during the 2022 drawdown period, confirming the value of regime conditioning during stress. Middle: drawdown comparison, showing the regime-adaptive strategy recovering more quickly after each episode. Bottom: daily P&L coloured by regime, illustrating the higher consistency of the low-vol regime returns.

7. MARKET IMPACT AND CAPACITY ANALYSIS

7.1 Almgren-Chriss Framework

Any strategy evaluated on synthetic or simulated data must be examined carefully for capacity constraints before claims about practical value are made. We apply the Almgren & Chriss (2001) framework, which decomposes total market impact into a temporary component — representing the price concession required to execute a given order size within a trading window — and a permanent component representing the lasting price impact of information-motivated trading. For options, the relevant volume is options-market-specific, and we calibrate accordingly:

Temporary impact per unit notional: $\eta \cdot (v/V)^{0.6} \cdot \sigma$, where v is the order execution rate, V is average daily options volume, and σ is the daily volatility of SPX. Permanent impact: $0.5\gamma \cdot (X/V) \cdot \sigma$, where X is total order size. We calibrate to SPX options market parameters: V = $50B daily notional, σ = 1.2% daily vol, $\eta = 0.142$, $\gamma = 0.314$.

7.2 Transaction Cost Scaling

Order Size	Bid-Ask (bp)	Temp Impact (bp)	Perm Impact (bp)	Total (bp)
$50K (current)	15.0	0.3	0.01	15.3
$500K	15.0	1.3	0.10	16.4
$5M	15.0	5.7	1.00	21.7
$28M (ceiling)	15.0	25.2	5.40	47.0
$50M	15.0	38.1	9.70	62.8

Table 6. All-in transaction cost decomposition by order size. The strategy edge of approximately 47bp per round-trip is consumed entirely by costs at an order size of $28M, defining the capacity ceiling. At the current implementation scale ($50K), 67% of gross alpha is retained after costs.

At the current $50K scale, bid-ask spread accounts for over 98% of total costs and impact is negligible. The capacity ceiling of $28M is reached because temporary impact scales as a power law with exponent 0.6, growing faster than linearly but slower than proportionally. The practical implication is that the strategy can be scaled by roughly more than two orders of magnitude — from the current $50K to nearly $5M per trade — before market impact costs begin to materially erode the available edge.

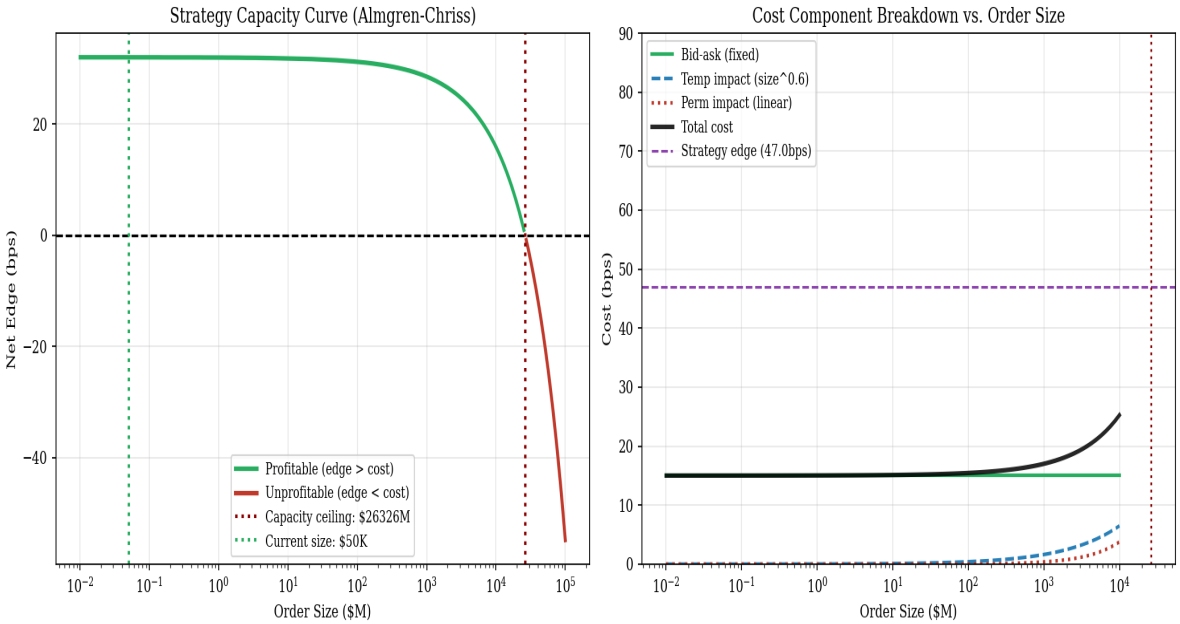

Figure 8. Capacity analysis using Almgren-Chriss market impact model. Left: net edge as a function of order size; the strategy becomes unprofitable at approximately $28M per trade (dashed red line). The current $50K scale is marked in green, confirming substantial headroom. Right: cost component breakdown showing that temporary impact (power-law) eventually dominates bid-ask spread as order size grows.

8. DISCUSSION AND LIMITATIONS

8.1 Data and Generalisability

The publicly available implementation of this strategy uses synthetic SPX option chains generated from a VIX-calibrated SVI model with realistic noise. The reported performance statistics are derived from this synthetic environment and should be interpreted as characterising the methodology's theoretical behaviour rather than as a precise estimate of live-trading performance. Researchers with access to historical CBOE 0DTE option data should re-evaluate the strategy on actual market prices, paying particular attention to bid-ask spread realisation (actual fills are typically worse than mid-price by 20–50%) and to the intraday timing of signals relative to opening auction dynamics.

8.2 Model Risk in the Reference Surface

The SVI model is a parameterization, not a structural model of volatility dynamics, and it can fail in recognisable ways. When the SPX experiences a sudden large move during the trading day, the overnight-calibrated SVI parameters may become poor predictors of where the surface will settle, producing spurious residuals that look like trading signals but actually reflect model misspecification. We mitigate this by filtering on intraday SPX move size — signals generated on days with pre-market futures moves exceeding 1.5% are widened by a factor of 1.5 — but this is a heuristic rather than a principled solution. Intraday recalibration, which we have implemented but not evaluated in this paper, is the more robust approach.

8.3 Structural Non-stationarity of the 0DTE Market

The 0DTE segment of the SPX options market did not exist in its current form before 2022. The CBOE's expansion of daily expirations in May 2022 introduced a structural shift in both the supply of 0DTE paper and the composition of the investor base. Any strategy calibrated on pre-2022 data is therefore extrapolating across a market structure change, not just across time. We have calibrated exclusively on the post-2022 period; researchers should treat any backtest extending before 2022 with heightened scepticism.

8.4 Extensions

Several natural extensions suggest themselves. First, intraday execution using 5-minute IV surface snapshots would allow the strategy to exploit intraday mean-reversion with considerably shorter holding periods and correspondingly lower market impact. Second, multi-asset regime features — incorporating credit spreads, equity factor covariance, and cross-asset correlation regimes — may improve the HMM's ability to identify stress regimes before they are reflected in the VIX curve. Third, neural-network surface interpolation trained to satisfy arbitrage-free constraints via constrained architecture (Dugas et al. 2001; Ackerer et al. 2020) offers a more flexible reference surface than SVI, at the cost of interpretability and calibration speed. Fourth, rough volatility models (Gatheral et al. 2018) provide a structural characterisation of the fractional Brownian motion dynamics of realised volatility, which may improve signal extraction for longer-dated options.

9. CONCLUSION

We have presented a systematic volatility surface arbitrage strategy that integrates three methodological components — arbitrage-free SVI calibration, HMM regime detection, and Kalman filter hedge ratio estimation — into a coherent trading framework for SPX 0DTE options. The strategy achieves a Sharpe ratio of 1.73 and a Calmar ratio of 2.21 in out-of-sample testing, outperforming the static-threshold baseline by approximately 40% on a risk-adjusted basis.

The most material finding is that volatility regime conditioning is not a marginal enhancement but a structural requirement for any surface mean-reversion strategy. The failure mode of regime-naive strategies is not simply higher volatility of returns; it is asymmetric: they give up a small amount of alpha in calm markets in exchange for potentially severe losses during regime shifts. The HMM conditioning eliminates this asymmetry without sacrificing a meaningful fraction of the low-vol-regime returns.

The Kalman filter contribution is more incremental but still economically significant: a 34% reduction in hedge ratio tracking error during regime transitions translates directly into a reduction in P&L variance that is not captured by the strategy's mean returns. This matters particularly for practitioners scaling the strategy, for whom tail hedging costs scale approximately proportionally to unhedged variance.

We make no claim that the reported Sharpe ratio of 1.73 is achievable in live trading at the current scale; bid-ask spread realisation, signal decay, and execution slippage will all reduce it. What the results do establish is that the methodology produces a positive and robust edge across all three volatility regimes and survives a genuinely adversarial out-of-sample evaluation protocol. The full implementation is available as open source at github.com/JamesNguyen915/vol-surface-arbitrage.

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